Module Review: Bacterial Mla Intermembrane Phospholipid Transport in *Pseudomonas putida* KT2440

Target: *Pseudomonas putida* KT2440 (PSEPK; NCBI taxon 160488; proteome UP000000556) **Module:** Mla intermembrane phospholipid transport (OM MlaA/VacJ – periplasmic MlaC – IM MlaFEDB) **Local bucket reviewed:** KEGG ppu02010 "ABC transporters" (151 primary genes; 207 candidates)

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1. Executive summary

The Mla system is **fully present and satisfiable** in *P. putida* KT2440. All conserved components map cleanly to the genome:

- **Inner-membrane MlaFEDB** + **periplasmic MlaC** are encoded by a single contiguous operon, **PP_0958–PP_0962** (`mlaF–mlaE–mlaD–mlaC–mlaB`), plus a Pseudomonas-specific BolA gene (PP_0963).
- **Outer-membrane MlaA/VacJ** is encoded by **PP_2163** (`vacJ`), an MlaA-family lipoprotein at an unlinked locus.

The two most curation-relevant problems are **hidden/mis-labeled subunits**, not gaps:

1. **MlaC** = **PP_0961** and **MlaB** = **PP_0962** are annotated only as generic *"toluene tolerance protein"* (`ttg2D`/`ttg2E`). Domain evidence (PF05494/IPR008869 for MlaC; PF13466 STAS for MlaB) **and KEGG Orthology (PP_0961→K07323 mlaC; PP_0962→K07122 mlaB, both single-copy)** firmly identify them. Without domain/KO-level review these steps would be **falsely scored as gaps**.
2. **MlaA/VacJ (PP_2163)** is a lipoprotein, not an ABC subunit, so it is **not in the ppu02010 candidate list**. Its absence from the ABC bucket is expected and must not be read as an OM gap.

A separate paralogous **Mce/MlaE-family ABC system (PP_0140–PP_0142)** is present in the candidate list and should **not** be counted as core Mla (over-annotation risk on PP_0140's generic "Mce/MlaD" label). Note that this Mce operon shares KEGG KOs **K02065/66/67 (mlaF/E/D)** with the true Mla operon, so KO-based scoring cannot distinguish them — the module must be anchored on the **Mla-specific KOs K07323 (mlaC)** + **K07122 (mlaB)** plus operon context.

Module step verdicts at a glance:

Step	Verdict	Gene(s)	Evidence tier
1. OM MlaA/VacJ	**covered**	PP_2163 (vacJ)	UniProt PF04333 + KO K04754; add to module (outside bucket)
2. Periplasmic MlaC	**covered**	PP_0961 (ttg2D)	PF05494 + **single-copy KO K07323**; re-annotate
3. IM MlaFEDB	**covered**	PP_0958/0959/0960 + MlaB PP_0962	PF00005/02405/02470 + KOs; MlaB via **single-copy KO K07122**
OM porin partner	candidate_uncertain	unassigned	low priority; not required for satisfiability
Paralogs to exclude	keep separate	PP_0140–0142 (Mce), PP_0599/PP_2577 (PqiB), PP_1737 (2nd MlaA)	distinct systems / uncertain

Evidence is a mix of **strong homology/domain assignment (curated UniProt + Pfam/InterPro, direct for KT2440 proteins)** and **functional transfer from other Gram-negatives** (*E. coli*, *P. aeruginosa*). Direct experimental phenotyping of the Mla/Ttg2 locus exists in *P. putida* strains (solvent tolerance), giving indirect functional support.

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2. Target-organism pathway definition

Process included: Protein-mediated exchange of glycerophospholipids (GPLs) across the cell envelope of a diderm (Gram-negative) bacterium, via three coupled sub-assemblies: (i) an OM MlaA/VacJ–porin interface that accesses GPLs at the OM inner/outer leaflet; (ii) a soluble periplasmic MlaC carrier that ferries a single GPL across the periplasm; (iii) an ATP-hydrolyzing IM ABC complex MlaFEDB. In *E. coli* the net direction is largely **retrograde** (OM→IM), restoring OM lipid asymmetry, though anterograde roles have been debated (P 36459067). The module is deliberately direction-neutral.

Neighboring pathways to keep separate (do not merge): - **Lpt LPS export** (KT2440: `lptB` PP_0953, `lptF` PP_0982, `lptG` PP_0983, plus LptA/C/D/E) — transports LPS, not GPLs; shares the ABC bucket but is a distinct module. - **Mce/Pqi-type intermembrane transporters** (KT2440: PP_0140–PP_0142) — architecturally related (MlaD/Mce and MlaE domains) but a separate system. - Generic **ABC import systems** in ppu02010 (amino acid, sugar, phosphonate, metal uptake) — unrelated to lipid trafficking. - Broad overview maps (KEGG "ABC transporters" ppu02010; "Transporters" BRITE) are collection buckets, not the Mla module.

Alternate names / database definitions: - Mla = "Maintenance of Lipid Asymmetry." - **Pseudomonas/lineage synonym:** **`ttg2`** ("toluene tolerance genes"). KT2440 uses `ttg2D` (=MlaC) and `ttg2E` (=MlaB); the DOT-T1E literature labels the operon `ttg2ABCDEF`. - MlaA is also called **VacJ** (KT2440 uses `vacJ`). - OM complex is "MlaA–OmpC/OmpF" in *E. coli*; the porin partner in *Pseudomonas* is not firmly established.

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3. Expected step model

#	Module step	Molecular player	Expected in KT2440?
1	OM phospholipid handling	MlaA/VacJ (+porin)	Yes
2	Periplasmic shuttle	MlaC carrier	Yes
3	IM ATP-coupled complex	MlaFEDB (F=ATPase, E=permease, D=MCE, B=STAS regulator)	Yes

MlaB (STAS domain) is a regulatory subunit of the IM complex; the canonical complement is "six Mla proteins, MlaFEDBCA" (P 33199922).

4. Candidate genes and evidence

Core Mla — high confidence (all direct KT2440 protein evidence via UniProt/Pfam)

Gene	Locus	UniProt	Assigned role	Key domain evidence	Status
mlaF	PP_0958	Q88P94	IM ABC ATPase	PF00005; "ABC transporter superfamily, **MlaF family**"	Covered
mlaE	PP_0959	Q88P93	IM permease (5 TM)	PF02405; MlaE permease family; IM multipass	Covered
mlaD	PP_0960	Q88P92	MCE substrate-binding subunit	PF02470; **IPR030970 (MlaD-specific)**	Covered
mlaC	**PP_0961**	Q88P91	**Periplasmic GPL carrier**	**PF05494 / IPR008869 (MlaC/Ttg2D)**; SignalP 1–22 → periplasm	Covered (re-annotate)
mlaB	**PP_0962**	Q88P90	**STAS regulatory subunit**	**PF13466 (STAS)**; 100 aa	Covered (re-annotate)
vacJ/ mlaA	PP_2163	Q88KX6	**OM lipoprotein**	PF04333 / IPR007428; **MlaA family**; lipoprotein signal 1–31	Covered (not in bucket)

Operon context: `kdsD` (PP_0957) – **mlaFEDCB** (PP_0958–0962) – **ttg2F/BolA** (PP_0963, PF01722). Contiguous synteny mirrors the *E. coli* `mlaFEDCB` operon and confirms co-assignment.

Independent KEGG-Orthology (KO) confirmation: PP_0958→**K02065 (mlaF)**, PP_0959→**K02066 (mlaE)**, PP_0960→**K02067 (mlaD)**, PP_0961→**K07323 (mlaC)**, PP_0962→**K07122 (mlaB)**, PP_2163→**K04754 (mlaA/vacJ)**. Crucially, KEGG assigns MlaC and MlaB to PP_0961/PP_0962 despite their generic "toluene tolerance protein" free-text names, and the two Mla-specific KOs — **K07323 (mlaC) and K07122 (mlaB) — are single-copy in *ppu*** and sit inside the operon. This gives high-confidence coverage of module step 2 (MlaC) and the MlaB regulatory subunit.

> **Curation caveat — shared KOs are NOT Mla-specific.** KEGG KOs K02065/K02066/K02067 (mlaF/E/D) are also assigned to the paralogous Mce operon (PP_0141→K02065, PP_0142→K02066, PP_0140→K02067). KO-based module scoring therefore cannot, on its own, distinguish the true Mla operon from the Mce system and could double-count the F/E/D subunits. **Anchor the Mla module on the Mla-specific single-copy KOs K07323 (mlaC) + K07122 (mlaB) plus operon context** — not on the shared mlaF/E/D KOs. The PqiB proteins PP_0599/PP_2577 (K06192, pqiB/letB) are a further distinct system.

Related but distinct — candidate_uncertain

Gene	Locus	UniProt	Note
PP_0140	PP_0140	Q88RI9	"Mce/MlaD domain" (PF02470, **generic IPR003399**, not MlaD-specific IPR030970). Part of a separate Mce/Pqi-like transporter.
PP_0141	PP_0141	Q88RI8	ABC ATPase (PF00005) of that system.
PP_0142	PP_0142	Q88RI7	Permease fusing **MlaE (PF02405)** + **STAS (PF13466)** domains — hallmark of Mce/Pqi-type, not core MlaE.

Not part of this module (present in the bucket, keep separate)

`lptB` (PP_0953), `lptF` (PP_0982), `lptG` (PP_0983) — LPS export; and all amino-acid/sugar/metal/phosphonate ABC importers listed among the 207 candidates.

Genome-wide paralog census (Pfam over proteome UP000000556)

Copy-number analysis both firms up the assignments and exposes the paralogs that must be excluded from the core Mla module:

Pfam	Family	# in proteome	Core Mla member	Other (non-Mla) members
PF05494	MlaC/Ttg2D	**1**	**PP_0961 (MlaC)**	none — single-copy, unambiguous
PF02405	MlaE permease	2	PP_0959 (MlaE)	PP_0142 (Mce-system permease, MlaE+STAS fusion)
PF02470	MlaD/Mce	4	PP_0960 (MlaD)	PP_0140 (Mce), PP_0599 & PP_2577 (PqiB-like)
PF13466	STAS	4	PP_0962 (MlaB, by operon context)	PP_4364, PP_2166 (anti-σ antagonists), PP_0142
PF04333	MlaA/VacJ	2	**PP_2163 (VacJ)**	**PP_1737 (paralog, uncertain)**

Take-aways: - **MlaC is single-copy → PP_0961 is a definitive assignment.** - KT2440 encodes **three architecturally related but functionally distinct intermembrane systems**: Mla (PP_0958–0963), a **Mce/Pqi-type ABC system** (PP_0140–0142), and a **PqiAB/PqiB system** (PP_0599, PP_2577). These belong in separate modules. - **A second MlaA/VacJ paralog, PP_1737** (248 aa, PF04333, mis-labeled "ABC transporter"), has **no detected lipoprotein signal** and lies next to a patatin/PNPLA phospholipase (PP_1736) and a lipid A acyltransferase (PP_1735) — a lipid-metabolism neighborhood distinct from the Mla operon. **PP_2163 (VacJ)**, which carries the canonical lipoprotein signal and the `vacJ` name, remains the primary OM Mla component; **PP_1737 should be `candidate_uncertain`**.

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5. Gaps, ambiguities, and likely over-annotations

- **No true gaps.** Every module step is encoded.
- **Hidden subunits (annotation gap, not sequence gap):** MlaC (PP_0961) and MlaB (PP_0962) carry only legacy `ttg2D` / `ttg2E` "toluene tolerance protein" names. This is the single biggest curation risk — automated step-scoring keyed on the string "Mla" would mark steps 2 and the MlaB part of step 3 as missing.
- **Bucket-scope artifact:** MlaA/VacJ (PP_2163) is correctly outside ppu02010 (it is not an ABC protein). Module scoring must look beyond the KEGG ABC bucket for step 1.

- **Over-annotation risk:** PP_0140's "Mce/MlaD domain-containing" label should not be promoted to core MlaD. The MlaD-specific signature (IPR030970) is on PP_0960, not PP_0140.
- **Porin partner unresolved:** the MlaA-associated OM porin (OmpC/F equivalent) is not assigned in KT2440; low priority for module satisfiability but relevant for a complete OM sub-complex.
- **PP_0963 (ttg2F, BolA family):** a Pseudomonas operon extension of undefined Mla function — ambiguous, likely accessory/regulatory.

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6. Module and GO-curation recommendations

Step verdicts for the generic module:

Module step	Verdict	Genes
1. OM MlaA/VacJ interface	**covered**	PP_2163 (add to module; outside candidate bucket)
2. Periplasmic MlaC shuttle	**covered**	PP_0961 (re-annotate MlaC)
3. IM MlaFEDB complex	**covered**	PP_0958 (F), PP_0959 (E), PP_0960 (D), PP_0962 (B)

Module document / boundary actions: - The generic module boundaries are **correct** for KT2440; no `module_needs_revision`. Recommend adding an explicit **`ttg2` synonym note** and mapping `ttg2D→MlaC`, `ttg2E→MlaB` so Pseudomonas loci are captured. - Add PP_2163 (VacJ) as the step-1 gene despite its exclusion from the KEGG ABC bucket; note that MlaA/VacJ is a lipoprotein, not an ABC subunit. - Mark **PP_0140–PP_0142 as `candidate_uncertain` / assign to a separate Mce–Pqi module**, not core Mla. - **GO curation:** the KT2440 UniProt entries for PP_0960/0961/0962 would benefit from explicit GO annotations `GO:0120010 (intermembrane phospholipid transfer activity)` / `GO:0032365 (intracellular lipid transport)` / `GO:0015221` context and the process term for OM lipid-asymmetry maintenance; PP_0961 should also get `phospholipid binding`. No new GO *term* requests appear necessary — existing Mla-related terms suffice.

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7. Genes to promote to full `fetch-gene` review

1. **PP_0961 (ttg2D → MlaC)** — highest priority: periplasmic carrier mis-labeled "toluene tolerance protein"; re-annotation directly satisfies module step 2.
2. **PP_0962 (ttg2E → MlaB)** — STAS regulatory subunit mis-labeled; completes the IM complex.
3. **PP_2163 (vacJ/MlaA)** — confirm lipoprotein processing and add to module as step 1.
4. **PP_0140** — resolve Mce/MlaD vs. core-MlaD ambiguity; assign to the correct (separate) Mce/Pqi system.
5. **PP_1737** — second MlaA/VacJ (PF04333) paralog; determine whether it is a functional MlaA or a distinct phospholipase-associated lipoprotein (currently `candidate_uncertain`).
6. **PP_0963 (ttg2F, BolA)** — determine whether it is a bona fide Mla accessory subunit.

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8. Key references

- Tang et al. 2021, *Nat Commun* — MlaFEDB structure; "comprises six Mla proteins, MlaFEDBCA."

  P 33199922
- Wotherspoon et al. 2024 — MlaC–MlaD complex; periplasmic shuttle definition.

  P 39080293
- MacRae et al. 2023 — Mla protein–protein interactions / MlaC deep mutational scanning.

  P 37100290
- Abellon-Ruiz 2023 — review of Mla transport directionality.

  P 36459067
- Low & Chng 2021 — OmpC-Mla mechanism review.

  P 34753108
- Ekiert et al. 2022 — MlaFEDB mechanism synthesis.

  P 35981415

- Kaur et al. 2023 — *P. aeruginosa* MlaA/VacJ OM lipoprotein phenotypes.

  P 37660742
- Kim et al. 1998 — *P. putida* GM73 `ttg` mutants; Ttg2 = ABC transporter, solvent tolerance.

  P 9658016
- Dutta et al. 2026 — computational analysis of MlaA/MlaB/MlaE/MlaF distinctive features.

  P 41047745

Evidence-strength notes

- **Direct for KT2440 proteins:** all Pfam/InterPro domain and subcellular-location assignments (UniProt UP000000556). This is homology-based annotation of the *actual* target-strain sequences — strong for identity, but the transport *function* is inferred.
- **Functional transfer (strong):** Mla mechanism from *E. coli*/*P. aeruginosa* — high sequence and architectural conservation; transfer to KT2440 is well justified.
- **Direct-ish phenotype in species (indirect for module):** *P. putida* `ttg2` solvent-tolerance mutants (GM73/DOT-T1E lineage) link this locus to envelope stress but do not by themselves prove GPL transport in KT2440.
- **Open experiments:** GPL-transport assay / OM-asymmetry phenotype of a KT2440 `mlaC` (PP_0961) or `mlaFEDB` deletion; identification of the MlaA porin partner; role of the BolA gene PP_0963.
